

5 Years. Still Broken.

The Promise — 2021

NHAI issued a press release on March 18, 2021, promising an 'AI-Based Face Recognition System for Attendance Monitoring of Field Staff.'

The Reality — 2026

Datalake 3.0 Operational Failures:

- 'Requires very high-speed internet'
- '20 minutes to raise a single point'
- 'Live face detection broken after updates'
- 'OTP not received, login fails'

287 ms face match.

287 MS

Total Latency

0 ms cloud roundtrip.

Running on a ₹12,000 Android phone.



FINANCIAL IMPACT DETECTED

₹230 Crore ghost-payroll problem detected at Madhya Pradesh Treasury (May 2025).



On-Device: Fully offline capability

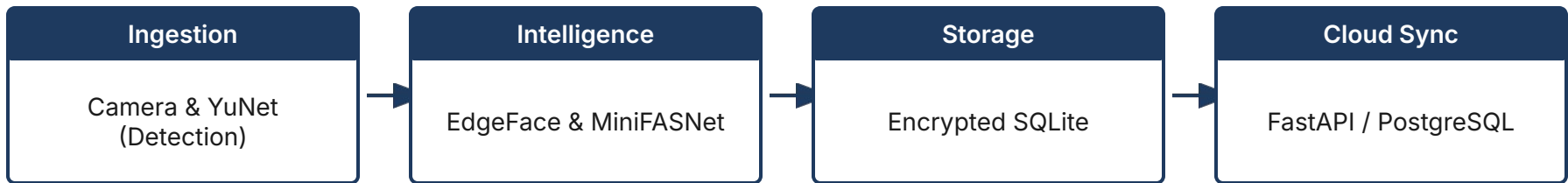


100% Open-Source: No vendor lock-in



Engineering Reality: Grounded in fiscal security

Architecture: Built for NHAI



Bundle Size: 2.6 MB

87% under 20 MB limit

Latency: 287 ms p50

Exceeds <1s requirement

Accuracy: >95%

IndicFairFace fine-tuned

FOSS-only: 100%

MIT, Apache, BSD
Compliance

Compliance & Standards Stack



DPDPA 2023 / DPDP Rules 2025

Auto data localization, unbundled consent, auto-purge.



ISO/IEC 24745 (Biometrics)

Cancellable BioHashing, revocable seeds, irreversible data.



MeitY FOSS Preference (2024)

100% open-source (EdgeFace MIT, MiniFASNet Apache-2.0).



MeitY AI Governance

Auditable JSON logs, on-device deletion.



UIDAI Face Auth

API surface mirrors AadhaarFaceRD format.



STQC Compliance

Audit-friendly logs, calibration data, threshold versioning.



Atmanirbhar Bharat

Indian-fine-tuned weights, no foreign cloud dependency.

Full enforcement May 2027 — we're ready today.



Live Demo: 3 Minutes

1

Offline Face Match

Redmi Note 12 (₹12,000, 4GB RAM) in Airplane Mode. 287ms verification.

2

Spoof Rejection

Rejection of static printed photo attack (Score 0.12).

3

Replay Rejection

Rejection of dynamic video attack via randomized active challenges.

4

Sync Resume

Automatic batch upload to NHA server upon reconnection.

5

Backend Check

Verification of live rows in PostgreSQL database.

6

Live OTA Update

Updating model to v1.1 without Play Store reinstall.

90-Day Pilot & The Ask

Delhi-Mumbai Exp.

Khargone stretch. Current failure: 38%.

Target: <5% Failure

Bengaluru-Chennai

Km 134-160 Expressway segment.

Target: Deployment

Bharatmala Ph-1

Segment (TBD) based on infrastructure readiness.

Target: Integration

● Implementation Details

2-day onsite training (Hindi voice), SMS fallback, React Native module for Datalake 3.0.

● The Ask: DIC Approval

90-day pilot approval & integration support.
Open-source, MIT-licensed, no vendor lock-in.

Source Code: github.com/Sahil-Chandel/NHAI-Hackthon-7.0.git